RESPONSE TO REVIEWER REPORTS FOR PHD THESIS

Abhirup Mukherjee (18IP014)

RESPONSE TO PROF. PINAKI MAJUMDAR

Question 1. Is there an unique impurity model, in terms of geometry, that describes the lattice? The thesis uses a 1 + 4 geometry, one can also imagine a 1 + 8 or 1 + 12 .. geometry.

Response. Strictly speaking, our approach does not pin down a unique impurity model given some lattice model. Nevertheless, a controlled hierarchy can indeed be designed for systematically increasing the accuracy of our procedure. Instead of increasing the number of correlated bath sites (1+4 to 1+8), the idea is to increase the number of correlated impurity sites. The cluster beyond 1+4 would be one where there are five impurity sites and the appropriate number of correlated bath sites surrounding them. The complexity of solving such an impurity model is of course higher. This has now been explained better in the thesis.

Question 2. For a particular choice of geometry there are more than two parameters. The geometry used in the thesis has an interaction JK and an attraction W . Are these uniquely determined by the U and t?

Response. Actually, the lattice model that our chosen auxiliary model maps on to is an extended Hubbard model that naturally involves a nearest-neighbour spin-exchange coupling J in additional to the terms in the "standard" Hubbard model. This coupling J uniquely determines the auxiliary model Kondo coupling J_K. The coupling W is however not uniquely determined by U. See the response to Q11 for more details.

Question 3. The impurity model is much more complicated than the usual single impurity Anderson model (SIAM). The URG is used to solve it. Is that an exact RG or is some truncation involved? The original SIAM was solved 'exactly' through numerical RG by Krishnamurthy, Wilkins and Wilson. Does the URG work at the same level?

Response. While the extended impurity model (eSIAM) is more complicated than the usual SIAM, it can be solved to a large extent using more "standard" approaches like numerical RG; obtaining specific quantities might be more easy or difficult depending on the specifics. Getting to the actual question, the unitary RG involves a hierarchy of contributions arising from a cluster expansion of the renormalised Hamiltonian into scattering vertices of various orders [1]. These describe scattering processes involving four electrons, six electrons and so on; such terms are generated during the RG flow. In practise, one has to truncate the expansion and restrict the RG equation to a finite order of vertices. In terms of concrete results, the unitary RG treatment of the single channel Kondo model [2] is able to capture all major numerical RG results, such as the flow to the strong coupling fixed point, thermodynamic quantities such as the impurity contributions to the magnetic susceptibility and specific heat, and the Wilson ratio. Similar benchmarking was also carried out for the two-channel Kondo model [3].

Question 4. The RG focuses on low energy physics, while the spectral function has high energy features as well. Methods like DMFT - more about which later - capture both the low energy and the Hubbard band features. What about the present method?

Response. The unitary *can* obtain high-energy features as well. This is done by using the renormalised couplings at various points along the RG flow. Renormalised Hamiltonians closer to the UV theory affect the high-frequency spectral features while those closer to the IR fixed point affect the low frequency features. By combining the results from various points along the RG flow, one can reconstruct dynamical quantities like the spectral function and self-energy as well as thermodynamic quantities [4, 5] such as the temperature dependence of the impurity susceptibility [2]. We are currently using such a strategy (as part of a more involved program) to reconstruct the finite temperature phase diagram of the particle-hole symmetric Hubbard model in the limit of infinite dimensions, and the results are quite promising.

Question 5. The lattice 'phase diagram' at half filling is just a line, with U/t as axis. It goes from a Slater insulator at small U/t to the Heisenberg-Mott insulator at large U/t. There is no room for a metal, pseudogapped or otherwise, at T = 0. While the flow diagram of Fig.5.4 is fascinating, what are the corresponding Hubbard parameters?

Response.

Question 6. Exactly which limitations of current impurity-based approaches motivate the present construction?

Response. A number of limitations of the current impurity-based methods were laid out in Sec 1.4 of the thesis. These include

- the inability of dynamical mean-field approximation-based methods to expose the underlying self-consistent impurity model Hamiltonian, often rendering the physics opaque,
- the sensitivity of static and dynamical quantities obtained from cluster methods to periodisation schemes,
- the increased computational cost required to obtain sufficient resolution in frequency, and
- the additional errors incurred in order to operate in phases with long-ranged correlations, which might be necessary to obtain spin liquid or other fractionalised phases that might be relevant to the physics of the cuprates and heavy fermions.

We have now addressed this point better within the thesis.

Question 7. However, for a reader the starting question would be what is the controlled approximation? Successful many-body methods are associated with a identifiable control parameter or asymptotic limit: for example, for the Wilson NRG it is the logarithmic energy-scale separation, for DMFT it is the locality of the self energy as $d \to \infty$, for tensor-network methods it is the low-entanglement structure, etc. What controls the URG for the extended impurity models?

Response. This was discussed in the response under Q3. The unitary RG involves a hierarchy of contributions arising from a cluster expansion of the renormalised Hamiltonian into scattering vertices of various orders [1]. These describe scattering processes involving four electrons, six electrons and so on; such terms are generated during the RG flow. In practise, one has to truncate the expansion and restrict the RG equation to a finite order of vertices

Question 8. One qualitative question: while the 1 + 4 cluster, for example, and the associated translation operators preserve the symmetry of the 2D lattice (modulo any long range order), does the process adopted generate the exact Greens functions of the lattice model?

Response. In principle, the process lays out a hierarchy of contributions to the Greens function that would capture the lattice Greens function exactly. In practise, we have restricted the calculations to the first few terms; this is supported by the local nature of the impurity correlation that helps the higher terms drop off. The method of accounting for the additional terms are shown in eqs. 4.24 and 4.25.

Question 9. Benchmarks: The 2D Hubbard model has been investigated using a variety of high-accuracy numerical methods, including auxiliary-field quantum Monte Carlo, tensor-network approaches, exact diagonalisation, dynamical cluster approximation, cellular DMFT, variational cluster approximation, etc. Benchmark comparisons in- volving quantities such as the ground-state energy, double occupancy, quasiparticle weight, spectral function, self-energy, and spin correlation would have helped establish the validity of the auxiliary Hamiltonian description.

Question 10. Uniqueness of the reconstruction: In principle, one could imagine alternative em- bedded clusters of different size or geometry while employing the same translation- operator formalism. For example, one might consider 1 + 4, 1 + 8, or larger auxiliary clusters. If the formalism defines a systematically controlled hierarchy, progressively larger clusters should converge towards identical lattice observables. Discussion of such a convergence hierarchy would be useful.

Response. A controlled hierarchy can indeed be designed for systematically increasing the accuracy of our procedure. Instead of increasing the number of correlated bath sites (1+4 to 1+8), the idea is to increase the number of correlated impurity sites. The cluster beyond 1+4 would be one where there are five impurity sites and the appropriate number of correlated bath sites surrounding them. The complexity of solving such an impurity model is of course higher. This has now been explained better in the thesis.

Question 11. Interpretation of the Auxiliary Phase Diagram: The RG flows reveal a remarkably rich fixed-point structure. Viewed as the phase diagram of the auxiliary Hamiltonian, these results constitute an interesting contribution to quantum impurity physics. The interpretation as the phase diagram of the square-lattice Hubbard model is less straightforward. The microscopic Hubbard model is specified by the parameters (t, U), whereas the auxiliary Hamiltonian is described by the larger parameter set (Ud , W, JK , V, . . .). Although the relation U = Ud − W connects the two descriptions, it does not uniquely determine the auxiliary Hamiltonian from the microscopic Hubbard interaction. The mapping from the Hubbard model to the auxiliary-cluster description appears non unique.

Response. Yes, it is true that the program laid out in the thesis does not map a lattice model to a unique impurity model. Nevertheless, we expect the qualitative features of the lattice model such as the nature of the phases and effective theories to be reconstructed through our tiling procedure. In other words, the primary goal of this approach is to discover the broad theories associated with lattice models of a certain kind (the extended Hubbard model, in this case) by mapping them to impurity models of the appropriate kind. As a final point, we would add that the appropriate impurity model is also determined by the phenomenology of the lattice model or the material that we are studying.

Question 12. Role of the Auxiliary Couplings: The phase diagram is organised in terms of auxiliary couplings such as W and J. These quantities determine the infrared fixed-point structure and therefore the physical interpretation of the various phases. Are uniquely determined by the microscopic Hubbard Hamiltonian?

Response. I believe the earlier response addresses this question.

Question 13. Relation to the half-filled Hubbard Model: The calculations are performed at half filling. For the nearest-neighbour square-lattice Hubbard model, long-range anti- ferromagnetic order is expected at zero temperature owing to perfect Fermi-surface nesting. The phase diagram presented here instead emphasises Fermi-liquid, pseudogap- metal and Mott-insulating fixed points. Has magnetic symmetry breaking been deliberately excluded? Similarly, the interpretation of the pseudogap phase requires clarification, since pseudogap behaviour is usually discussed in the context of doped cuprates rather than the half-filled nearest-neighbour Hubbard model.

Response. Spontaneous symmetry breaking has indeed been deliberately excluded, because the unitary RG method preserves all symmetries of the problem. In an earlier work involving the URG (not involving me), the potential for SSB in the system had been investigated by allowing for a small symmetry breaking field in the microscopic Hamiltonian and then checking its RG flow. Such an investigation has not been carried out in the present set of works. As a result, we only obtain symmetry-preserved insulators.

With regards to the pseudogap at half-filling, we note that non-Fermi liquids with partially-gapped Fermi surfaces have been observed within the nearest-neighbour Hubbard model *at half-filling* from DCA [6] and ΣDDMC [7], albeit at finite temperatures above the Mott insulating ground state. Our auxiliary model approach allows us to study the effects of such phases at zero temperature, by tuning additional couplings. Simulating our approach at finite temperatures should presumable lead to similar finite temperature phase diagrams; more work is needed in that regard.

Question 14. Maybe one could have had simpler notation for the electronic operators, I struggled to figure out what $c_{Z,\sigma,f}$ might be.

Response. We have now explained the notation in more detail within the thesis.

Question 15. The standard model for heavy fermions is the Anderson lattice. If eqn3.2 is the lattice model here, I guess the Anderson lattice corresponds to Ud = 0, W = 0 and J = 0. Yet the phase diagrams are plotted in terms of these parameters. How do we connect to Anderson lattice results?

Response. We first note that the phase diagrams are plotted as a function of $J_\perp$ (the inter-orbital Kondo coupling) and W_f (the $f-$layer bath correlation). The bath correlation for the conduction layer is set to zero (to make it only weakly interacting, compared to the $f-$layer). The phase diagrams are then plotted as a function of $J_\perp$ and W_f, to show how the behaviour changes upon tuning (i) the inter-layer connection, and (ii) the correlation of the $f-$layer. The Kondo lattice problem resides at large negative W_f, because that's when the electrons of the $f-$layer form local moments. The presence of the coupling $J_\perp$ implies that we are dealing with an extended periodic Anderson mode where the inter-orbital connectivity arises not just from one particle hybridisation but also from spin-exchange processes.

Question 16. There is detailed spectral work by D. E. Logan and coworkers on DMFT of the Anderson lattice using a local moment approximation. Do the spectral results here have any correspondence with them?

Response. While Logan and coworkers have a series of works on the periodic Anderson model, we focus on [8], where they deal with the particle-hole symmetric case that is pertinent to this thesis. The results obtained here agree qualitatively with those obtained in [8], in terms of the opening of the gap and the movement of the Hubbard satellite peaks; more work is necessary in order to compare quantitative claims such as scaling of certain quantities as a function of frequency.

RESPONSE TO PROF. SUBHENDRA D. MAHANTI

Question 17. What is the physical difference between "Insulator with spin mixing between impurity spin and bath spin" and "Mott Insulator". Explain

Question 18. What is the physics of charge isospin Kondo coupling K between the impurity and the bath: needs to be explained better.

Response. We first note that the effects of a charge isospin Kondo coupling hase not been explored within the thesis, and have only been mentioned in passing.

While the usual Kondo effect describes the screening of an impurity magnetic moment by the local spin density of the conduction bath, one can also envisage a scenario where it is the charge degrees of freedom of the impurity that is screened by the bath [9]. This can happen, for example, if the local correlation on the impurity site is negative. That makes the $n = 0$ and $n = 2$ configurations (unoccupied and doubly occupied) lower in energy. One can then define an *isospin* from this doublet of states, as well as a similar isospin for the doublet of states in the bath. Isospin-flip fluctuations will then involve transitions where an unoccupied state on the impurity scatters to a double occupied state while the bath simultaneously scatters in the opposite manner. It is worth noting that during the Schrieffer-Wolff transformation from the Anderson impurity model to the Kondo model, such charge Kondo terms are also generated but they reside in the high-energy part of the effective theory [10].

Question 19. I have a problem with chapter 4 section 4.2.2, Periodization of Auxiliary Model into Extended Hubbard Model using the lattice translation operator on the impurity cluster model. The operator $c_{r_d,\sigma}$ is a destruction operator for an impurity electron located at the lattice site r_d (originally site 0 of the impurity model). Also, in the impurity model there is a destruction operator associated with bath electrons associated with the same site, $c_{0,\sigma}$. There are N such bath destruction operators for N lattice sites. The number of fermion destruction operators is therefore N+1. When one translates the impurity over the N lattice sites there are 2N fermion destruction operators, not N as given in equations 2.6 and 2.7. What am I missing. I would have expected a 2-band extended Hubbard model, one for the "impurity" and the other for the bath.

Response. If the total number of lattice sites is N, then the number of bath operators is $N - 1$ because the impurity site is "embedded" in the lattice and forms the N^{th} operator. This is necessary in order for the impurity couplings to be sensitive to the lattice details. Applying translation operators does not double the number of operators because the same operators transform into one another: $c(r) \to c(r + a), c(r + a) \to c(r + 2a) \ldots c(r + Na) \to c(r)$.

Question 20. All the discussions are associated with an extended 2D Hubbard model. Are you assuming that the solution of this Hamiltonian is simply the periodic translation of the solution of the impurity embedded in a 2D lattice. Is this an ansatz or are there some justifications for this approach?

Response. Indeed, there's a set of arguments that relate the impurity model to the extended Hubbard model on the 2D square lattice. First, we show that the lattice model can be obtained by repeated translation of the impurity model (eq. 2.11). This motivates a form of the eigenstates of the lattice model in terms of those of the impurity model (eq. 2.20), via a many-body Blochs's theorem. Combining these two properties, we show that the lattice model Greens function can be expressed in terms of auxiliary mode quantities (eq. 4.23). This final relation is what allows us to reconstruct the low-energy physics of the lattice model.

Question 21. The novel feature is the condition of sufficiently large attractive W. It is not clear to me how such an attractive interaction appears in a simple 2D Hubbard model. But the most interesting observation is the appearance of a third fixed point, pseudo gap fixed point in a finite range of W.

Response. We feel that the attractive bath interaction W is best not understood as a microscopic coupling, but rather as an emergent effective coupling. It captures the tendency of the conduction electrons to favour unoccupied or double occupied states (relative to the singly-occupied states). Perhaps in a more illuminating sense, just as the Coulomb repulsion U represents the tendency that competes with the hopping processes in the simple Hubbard model, the bath correlation W represents the tendency that competes with the local Fermi liquid excitations that propagate out of the impurity site. In this sense, it is a proxy for the hubbard U in the neighbourhood of the impurity site; both compete against delocalisation processes.

Question 22. Is the pseudo gap fixed point associated with insulating and spin mixing as discussed in the previous chapter. Can you describe the spin density distribution and spin-spin correlation function associated with this fixed point. What role does the lattice symmetry play in the nature of this pseudo gap fixed point. What will happen if the 2D lattice is triangular or hexagonal?

Response. The unitary RG, as a matter of principle, does not allow for spontaneous symmetry breaking (SSB) because unitary transformations preserve all symmetries. One way of searching for SSB phases within such a formalism is to put in an explicit symmetry breaking term in the Hamiltonian and then study whether it is relevant under RG flow. We haven't carried out such an analysis within this thesis, and as a result, we don't find any such phase. Accordingly, in the absence of any magnetic order, there is no spin density in the system in any phase. The spin-spin correlation, however, shows long-ranged behaviour in the pseudogap phase (Fig. 5.13).

With regards to the role of lattice symmetry, we haven't checked whether our conclusions hold in other dimensions or other lattices. It sounds reasonable that changing the lattice symmetries will affect many of the results. However, a recently growing body of work [11–13] suggests that the result that the Hatsugai-Kohmoto model is the parent metal of the Mott insulator can be grounded on more robust ideas such as the breaking of generalised symmetries and anomalies of the Luttinger-Ward functional. These indicate that our results are likely to extend beyond just the 2D square lattice.

Question 23. I had some problem with the strong conclusion "Thus, the drastic change in nature of the real-space excitations- from locally well-defined Landau quasiparticles of the FL to the increasingly nonlocal excitations of the NFL Fermi arcs in the PG phase-appears to be dictated by a topological principle and, therefore, robust under renormalization. This signals the NFL Fermi arcs of the PG as an emergent phase of strongly interacting quantum matter- which we dub the Mott metal- that is the parent metal of the MI." What is the role of lattice dimensionality (D=2 vs 3) and lattice geometry (square vs triangular or Kagome') on this conclusion?

Response. As mentioned in the previous response, very strictly speaking our results apply only to the case of the 2D square lattice. However, a recently growing body of work [11–13] suggests that the result that the Hatsugai-Kohmoto model is the parent metal of the Mott insulator can be grounded on more robust ideas such as the breaking of generalised symmetries and anomalies of the Luttinger-Ward functional. These indicate that our results are likely to extend beyond just the 2D square lattice.

Question 24. What happens for large W_f and one increases $J_\perp$?

Response. For large W_f, the $f-$layer becomes a Mott insulator. The pseudogap phase shrinks to a smaller region due to the lack of non-Fermi liquid behaviour in the $f-$layer. Increasing $J_\perp$ brings about a Kondo insulator.

Question 25. He thinks the appearance of long-range correlation at the transition from Fermi liquid to a different phase might be relevant to the observation of strange metallic behavior in heavy-fermion systems at high temperatures above the quantum critical point. But all the calculations are at T=0K.

Response. While it is true that our results are all at $T = 0$, it stands to reason that some of these phases will have finite-temperature effects. For example, the pseudogap non-Fermi liquid behaviour we see near the transition arises from the quantum critical fluctuations; these fluctuations can survive up to certain temperatures and have finite temperature counterparts.

I. ADDITIONAL CHANGES

We have made the following additional changes to the manuscript:

- We have improved the inset figure of panel (a) in Fig. 4 for the impurity magnetisation (computed earlier for a smaller system size (49 x 49)) to that for a larger system size (77 x 77, same as the rest of the manuscript). The updated plot shows more clearly the dramatic rise of the impurity magnetisation towards the end of the pseudogap phase, and upon approaching the Mott critical point. This highlights the near complete breakdown of Kondo screening at Mott criticality as observed by us.
- We have added references to two works ([14, 15]) from the past that adopt an impurity model-based approach towards heavy fermions that are philosophically similar to ours.
- We have also fixed a typographical error below eq. (36) regarding the definition of the operator $r_{q\sigma}$.

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